

# Ahsanul Hoque

Full-Stack Software Engineer | Founder and CEO, Livora

Chattogram, Bangladesh (Remote) | ahsanulhoque22701048@gmail.com | linkedin.com/in/ahsanul-hoque-a31a3235b | github.com/AhsanulHoque22

## SUMMARY

Computer Science undergraduate (University of Chittagong, expected 2026) and Founder/CEO of Livora, a full-stack AI-powered healthcare platform that won 2nd place and an incubation package at the ITBI Student Startup Pitch Fest 2026 (CUET). Ships production-grade full-stack web applications end-to-end: React/TypeScript front-ends, Node.js/Express REST APIs, SQL data modeling, and security-conscious backend design (RBAC, rate limiting, prompt-injection defense). Integrates LLMs and AI features directly into shipped products (RAG, voice-to-text, OCR) and is currently fine-tuning an open-weight LLM in an active Kaggle AI hackathon. Comfortable working independently in distributed/remote settings under real deadlines.

## EXPERIENCE

**Founder and CEO - Livora (Self-employed)** | May 2026 - Present

*Chattogram, Bangladesh (Hybrid)*

- Lead product, engineering, and business strategy for Livora, an AI-powered healthcare platform serving patients, doctors, lab technicians, and hospital administrators
- Won 2nd place at the ITBI Student Startup Pitch Fest 2026 (CUET), securing 6 months of free incubation space, mentoring, and startup-ecosystem support worth BDT 100,000
- Conducted an AI training workshop on the platform for practicing surgeons in Chittagong

## SKILLS

**Languages:** Python, C++, TypeScript, JavaScript, SQL

**Front-End:** React, TypeScript, Tailwind CSS, Framer Motion, TanStack Query

**Back-End:** Node.js, Express, REST APIs, Prisma, Sequelize, Socket.IO, JWT/RBAC Auth

**AI / LLM:** LLM Integration (Gemini, Groq LLaMA), RAG, LoRA/QLoRA Fine-Tuning, Prompt Injection Defense, Langfuse, OCR (Tesseract), Speech-to-Text (Deepgram)

**Data, Infra and DevOps:** PostgreSQL, MySQL, Redis, Docker, CI/CD (GitHub Actions), pytest, Cloudinary, Pino, Sentry

**Other:** Data Structures and Algorithms, Competitive Programming (Codeforces, AtCoder, CSES), Embedded Systems (ESP32), Project Management

## PROJECTS

**Livora - AI-Powered Healthcare Platform** | Oct 2025 - Present

*React, TypeScript, Node.js, PostgreSQL, LLM/RAG, Socket.IO*

- Building a full-stack clinical platform covering appointment booking, voice-driven prescriptions, lab order/result management with payments, DICOM/FHIR and HL7 ingestion, and multi-disciplinary team (MDT) video consultations
- Implemented a persistent AI memory system and a RAG-based health assistant with medicine safety checks, risk stratification, and prompt-injection defenses; instrumented with Langfuse and Sentry

**Nascenia AI Hackathon - Bengali Medical Dialogue Generation** | Aug 2026

*Python, Transformers, LoRA/QLoRA, PEFT, Kaggle*

- Competing in a Kaggle-based national AI hackathon to fine-tune a  $\leq 3B$ -parameter LLM for Bengali doctor-patient dialogue generation under a strict inference-time parameter cap
- Built a local evaluation harness (BERTScore, Token-F1, ROUGE-L) matching the competition's exact scoring formula, and a config-driven LoRA/QLoRA fine-tuning pipeline with reproducible experiment tracking

**Aurora Gadgets - Full-Stack E-Commerce Platform** | Apr 2026 - May 2026

*React, TypeScript, Express, Prisma, MySQL, Docker*

- Built a complete storefront and admin system covering products, categories, brands, cart, and checkout with category-based product attributes
- Containerized the application with Docker for reproducible deployment

**Scraper Pipeline - AI-Powered Web Scraping and Extraction System** | Jul 2026 - Present

*Python, Playwright, Anthropic/OpenAI/Gemini/Groq/DeepSeek, pytest*

- Built a 4-stage pipeline (discover, collect, extract, clean) with a cost-aware multi-tier LLM cascade and cross-provider fallback chain that only escalates to a costlier model when deterministic validation of the cheaper tier's output fails
- Implemented Cloudflare/WAF bypass via headless-browser fallback, crash-safe atomic writes, resume-by-default batch processing, and a 92-test no-network suite covering every pure-logic component

### **Second Brain - Autonomous AI Agent for Productivity and Automation** | May 2026 - Present

*Claude Code, Python, Telegram Bot API, Git, Cron*

- Built an autonomous AI agent that runs study, startup, and daily-task workflows from a git-versioned Obsidian vault, with a Telegram bot as a mobile interface for logging progress, marking tasks done, and queuing background jobs
- Orchestrated 4 scheduled cron jobs (morning briefing, logging nudge, nightly re-planning, overnight task runner) atop a compiled knowledge-caching layer that processes source material once and serves every later query from cache

### **AgroEdge AI - Precision Agriculture Platform** | Jan 2026 - Apr 2026

*Python, Flask, Raspberry Pi, ThingSpeak, GitHub Actions*

- Built an AI-first precision-agriculture stack: data pipeline, irrigation model training, edge inference, a Flask web dashboard, and CI (GitHub Actions) running verification checks on every push
- Exposed model predictions via a documented REST API (OpenAPI 3.0) and integrated real-time sensor telemetry through ThingSpeak

### **Sales Data Warehouse and BI Reporting** | Aug 2025 - Sep 2025

*SQL, Python, Jupyter Notebook*

- Designed and built a modern SQL data warehouse from scratch, consolidating sales data via ETL and dimensional modeling
- Built interactive BI dashboards with Python (Pandas, Matplotlib, Seaborn) for trend and segmentation analysis

### **Warehouse Management and Billing System** | Freelance

*Node.js, MySQL*

- Built an inventory, billing, and sales-analytics system for a retail client

## **EDUCATION**

**University of Chittagong - B.Sc. in Computer Science and Engineering** | 2022 - 2026 (Expected)

**Govt. City College, Chattogram - Higher Secondary Certificate, Science (GPA 5.00)** | 2019 - 2022

**Chittagong Collegiate School - Secondary School Certificate, Science (GPA 5.00)** | 2013 - 2019

## **AWARDS AND ACHIEVEMENTS**

- 2nd Place, ITBI Student Startup Pitch Fest 2026, CUET - for Livora (incl. incubation support)
- 1st Place, Robo Soccer, EEE Fest 2025, University of Chittagong
- Runner-up, Hackathon Segment, Chittagong Science Carnival (hosted by CUSS)
- 1st Runner-up, Robo Soccer, Bangabandhu Innovation Fair 2024, Premier University
- Finalist, Bangabandhu Innovation Grant (BIG), 2023
- Champion, ETE DecaHz - SDG-focused idea competition (algae-based circular economy model)
- 2nd Runner-up, Intra-Department Programming Contest, 22nd Batch

## **LEADERSHIP AND ACTIVITIES**

- President, N-Cube (Science Club), Chittagong Collegiate School
- Joint Secretary, Debating Society, Chittagong Collegiate School